Statment of Purpose - Shaily Bhatt

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Language technologies are drastically altering access to information, resources, and opportunities. For example, technologies like translation between over hundred languages, information retrieval over thousands of documents, dialog agents, have all commendably advanced and pervaded our lives. However, despite impressive results on benchmarks, these systems are often brittle, have limited socio-cultural understanding, do not work well for low-resource languages, and perpetuate harmful societal biases. Further, insufficient evaluation and lack of interpretability of these black-boxes make them undependable. I am interested in developing reliable and inclusive language technologies. Particularly, I want to build better performing, socially-aware and fairer multilingual models, by both leveraging and increasing our understanding of current systems. Following my work on fairness at Google Research and multilingual evaluation at Microsoft Research, I now wish to develop:

1. Socially-grounded frameworks for holistic evaluation, fairness, and interpretability of NLP models.
2. Reliable and socially-cognizant models using insights from our current understanding of them.

I am confident that [redacted] would be an ideal fit for me. I now detail my work and future interests:

Holistic Evaluation of Multilingual Systems

Understanding the flaws of a system is the first important step towards improving them. As a Research Intern at Microsoft Research, I conducted a literature survey to understand the limitations of Language Models (LMs). We found that LMs were under-evaluated on low-resource languages and language varieties from under-privileged geographies and demographies. This survey, presented at **ICON 2021** [1], tremendously enhanced my grasp on the current landscape of NLP, especially the limitations of evaluation approaches. I realized that standard benchmark-style evaluation is prohibitively resource-intensive to scale to 100+ languages and their countless varieties. Moreover, these benchmarks do not point out specific bugs in the system and models can achieve high accuracy on them using spurious patterns. Thus, I turned to CheckList [5] which facilitates creation of test sets for specific capabilities using fill-in-the-blank templates. I worked on developing an algorithm to create multilingual CheckLists automatically and with humans-in-loop. I spearheaded experiments in 10 typologically diverse languages to show the efficacy of the technique. I also proposed evaluation metrics across multiple dimensions to illustrate the trade-offs amongst manual and automatic approaches of CheckList creation. Through this project, I learned the importance of critically analyzing not just the methods, but also the metrics for a task. This work was published at the **Findings of AACL 2022** [2]. I also built a working prototype of this algorithm which was used by product teams at Microsoft for evaluating their models.

While working on multilingual evaluation, I noticed a lack of holistic evaluation beyond typical NLP tasks. Particularly, evaluation for societal biases in non-western and multilingual contexts has been limited. Most NLP fairness research is in English and reliant on data, values, and justice systems of the West. As a Pre-Doctoral Researcher at Google Research, I have been working on bias evaluation in the Indian context. In my first-author work, published at **AACL 2022** [3], we present a research agenda to recontextualise fairness research in the Indian context, build data resources for India-specific fairness evaluation, and empirically demonstrate societal biases encoded in NLP corpora and models. I have also scaled these evaluations across Indian languages. During this project, I have come to realize that identifying societal-biases is inherently subjective, similar to tasks such as identifying hate speech or abusive language. Hence, obtaining and calibrating datasets for such tasks, in a way that respects diverse viewpoints is tricky. I have developed important insights on handling these complex subjective annotations, such as avoiding to take majority vote and going beyond quantitative measures and looking for qualitative insights. Moreover, being exposed to literature in fairness and ethics, especially the interplay of power structures, technology and historically minoritized communities, have greatly shaped my personal values. DEI efforts, such as co-organizing the Widening NLP Workshop and mentoring rising women researchers, are now even more central to my long term career goals.

Improving NLP Systems using Insights from Evaluation

Having worked on various forms of evaluation, mitigating these failure points is the next logical step in a system development cycle. At Microsoft Research, I improved a production scale hate-speech detection system using insights from the CheckList evaluation. I also led a case-study of practical efficacy of human-in-loop evaluation and improvement of systems in production settings. This resulted in presenting important open questions, like determining the right metrics, evaluation set quality, and optimal resource allocation. The work was published at the **Workshop on Human Evaluation at EACL 2021** [4]. Further, I developed important engineering skills through this project in working with large-scale deployment and latency constraints.

Similarly, in my current work at Google Research, following up on bias analysis, I am working towards mitigating India-specific biases in LMs. During the bias evaluation work, I realized that a lot of the societal

biases creep into LMs due to skewed or missing data. For example, discourse by and about marginalized communities is often completely missing or heavily polarized because of disparate literacy rates and technological access. Thus, I am leading a project to mitigate representational biases in LMs by correcting such data skews. I am working on a generalizable recipe for creating and using multilingual counterfactuals to mitigate societal biases. To create counterfactuals at scale, my idea is to leverage generative models like PaLM in collaboration with humans. This work has allowed me to work with exciting large scale models like PaLM and further explore human-model collaboration for the task of creating counterfactuals.

Future Interests, Graduate School, and Beyond

I believe that improved understanding of current systems and model development should progress hand-in-hand for developing reliable, responsible, inclusive, and culturally-aware NLP systems.

The definitions and expectations about reliability, performance, fairness, and explainability can greatly differ for users with varied demographics, literacy rates, and technological awareness. Having worked on re-contextualizing fairness research to the Indian context, I have seen these differences first hand. I believe analyzing whether and how systems incorporate these varying cultural expectations is imperative. Moreover language representations tend to be anglo-centric. For example, recent work showed that LMs do not possess non-western common-sense facts, such as colour of a bride's dress in some cultures of India being red instead of white [6]. Such failures are caused by data skews and non-representative test sets. Evaluation and analysis that can take such diverse and cultural viewpoints into consideration are necessary. Hence, to improve our current understanding of NLP models, I want to work on socially grounded frameworks for evaluating and analyzing NLP systems. Moreover, I would like to build general-purpose tools for this that they can be easily adapted to languages and dimensions, and scale with model size so that they are useful in practice.

Further, I want to leverage better understanding of NLP models to actively and iteratively improve them. For example, models ranging from BERT to GPT-3 have been shown to struggle with negation. The phenomenon gets more interesting in agglutinative languages like Swahili, where negated words are formed by adding prefixes to the root word. We observed during our multilingual CheckList work that negation for Swahili failed terribly, primarily because sub-word tokenization led to only the prefix being used in representation [2]. Algorithmic improvements taking such insights into account can dramatically improve the performance of models for low-resource languages. In this way, I wish to establish a feedback loop between understanding and building models to integrate insights from evaluation to create better and inclusive models.

Finally, I want to work on making NLP systems socially-cognizant. I believe this is important to building inclusive language technologies, so that models correctly recognize and contextually use culture-specific facts such as bridal colors. One of the ideas I have to this end is to incorporate commonsense or knowledge in NLP models in a modular manner. This should be done in a way that allows the model to use socially-relevant facts for a particular context in addition to other non-contextual facts. Moreover, modularity would also allow for easy updation of knowledge, thus allowing the model to adapt to the ever-changing world.

Based on my experience and interests, I believe I can contribute productively to the research in NLP at [redacted] My interests align closely with [redacted] I am also drawn by [redacted], and would like to explore this direction in limited data settings to create more inclusive systems. I find [redacted] relevant to my interests. I would also be excited to work with [redacted] and [redacted]. In the long run, I see myself working at the intersection of society, AI, and language, and on developing social good applications. The interdisciplinary environment and excellent course offerings at [redacted] would be ideal for developing the necessary skills for making an impact in these topics. [redacted] is known for its collaborative and thriving student community to which I can contribute positively. In short, PhD in CS from [redacted] at [redacted] would be the perfect step for me towards an academic career.

References

[1] **Shaily Bhatt** et. al., "On the Universality of Deep Contextual Language Models", *ICON 2021*

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[3] **Shaily Bhatt**, et. al., "Re-contextualizing Fairness in NLP: The Case of India", *AACL 2022*

[4] **Shaily Bhatt**, et. al., "A Case Study of Efficacy and Challenges in Practical Human-in-Loop Evaluation of NLP Systems Using Checklist", *Human Evaluation of NLP Systems Workshop at EACL, 2021*

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